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Question 1: IRIS Dataset

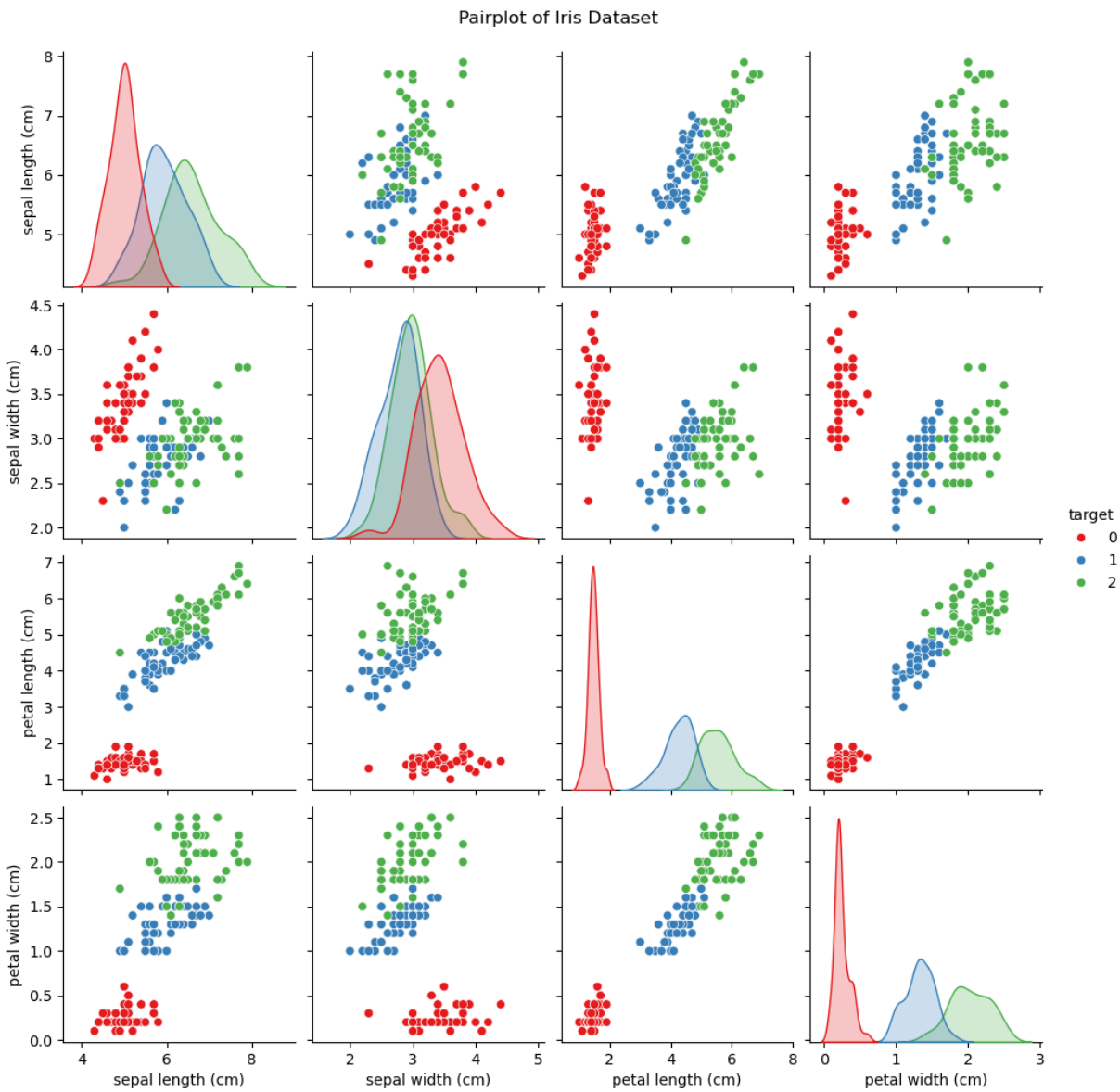
Classify the dataset IRIS using Decision tree. For dataset 'IRIS' consider correlation matrix for feature selection and compare between 4 models(all feature, only petals, only sepals, and best 3 features from correlation matrix). draw a bar chart of this summary. (use entropy as the criteria for classification).

General Requirements: a) perform all the initial EDA analysis b) Plot pair plot for IRIS. c) Remove Outlier in all the cases

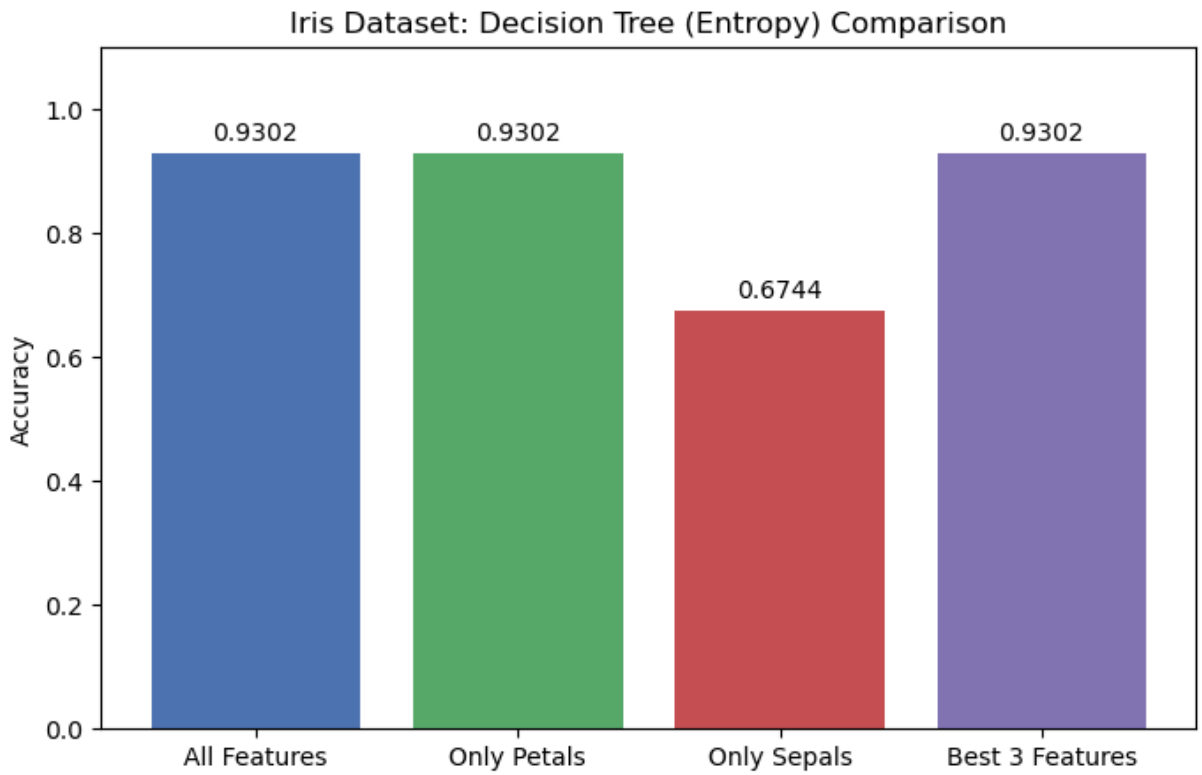
In [1]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris, load_wine, load_digits
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
from sklearn.ensemble import IsolationForest
from sklearn.decomposition import PCA
import warnings
warnings.filterwarnings('ignore')

# Common function to remove outliers using Isolation Forest
# (Robust for high dimensional and varied datasets without losing too much data)
def remove_outliers(df, target_col):
    X = df.drop(columns=[target_col])
    y = df[target_col]
    iso = IsolationForest(contamination=0.05, random_state=42)
    yhat = iso.fit_predict(X)
    mask = yhat != -1
    return df[mask].reset_index(drop=True)

print("=== IRIS DATASET ===")
iris = load_iris()
df_iris = pd.DataFrame(iris.data, columns=iris.feature_names)
df_iris['target'] = iris.target

# a) Initial EDA analysis
print("\n--- Initial EDA: Info ---")
print(df_iris.info())
print("\n--- Initial EDA: Describe ---")
print(df_iris.describe())

# b) Plot pair plot for IRIS
```

```
sns.pairplot(df_iris, hue='target', palette='Set1')
plt.suptitle("Pairplot of Iris Dataset", y=1.02)
plt.show()

# c) Remove Outliers
df_iris_clean = remove_outliers(df_iris, 'target')
print(f"\nOriginal Iris shape: {df_iris.shape}, After Outlier Removal: {df_iris_cle

# Feature Selection using Correlation Matrix
corr_matrix = df_iris_clean.corr()
print("\nCorrelation Matrix:\n", corr_matrix)

# Best 3 features based on absolute correlation with target
target_corr = corr_matrix['target'].drop('target').abs().sort_values(ascending=Fals
best_3_features = target_corr.head(3).index.tolist()

X_iris = df_iris_clean.drop(columns=['target'])
y_iris = df_iris_clean['target']

# Define models
all_features = X_iris.columns.tolist()
petal_features = [col for col in all_features if 'petal' in col]
sepal_features = [col for col in all_features if 'sepal' in col]

feature_sets = {
    'All Features': all_features,
    'Only Petals': petal_features,
    'Only Sepals': sepal_features,
    'Best 3 Features': best_3_features
}

accuracies_iris = {}
for name, features in feature_sets.items():
    X_subset = X_iris[features]
    X_train, X_test, y_train, y_test = train_test_split(X_subset, y_iris, test_size
    # Entropy as criteria
    clf = DecisionTreeClassifier(criterion='entropy', random_state=42)
    clf.fit(X_train, y_train)
    accuracies_iris[name] = accuracy_score(y_test, clf.predict(X_test))

# Draw a bar chart of this summary
plt.figure(figsize=(8, 5))
bars = plt.bar(accuracies_iris.keys(), accuracies_iris.values(), color=['#4C72B0',
plt.ylabel('Accuracy')
plt.title('Iris Dataset: Decision Tree (Entropy) Comparison')
plt.ylim(0, 1.1)
for bar in bars:
    yval = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2, yval + 0.02, round(yval, 4), ha='cent
plt.show()
```

```
=== IRIS DATASET ===

--- Initial EDA: Info ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   sepal length (cm)  150 non-null    float64
 1   sepal width (cm)   150 non-null    float64
 2   petal length (cm)  150 non-null    float64
 3   petal width (cm)   150 non-null    float64
 4   target             150 non-null    int64
dtypes: float64(4), int64(1)
memory usage: 6.0 KB
None

--- Initial EDA: Describe ---
       sepal length (cm)  sepal width (cm)  petal length (cm)  \
count         150.000000        150.000000         150.000000
mean            5.843333          3.057333           3.758000
std             0.828066          0.435866           1.765298
min             4.300000          2.000000           1.000000
25%             5.100000          2.800000           1.600000
50%             5.800000          3.000000           4.350000
75%             6.400000          3.300000           5.100000
max             7.900000          4.400000           6.900000

       petal width (cm)      target
count        150.000000  150.000000
mean           1.199333    1.000000
std            0.762238    0.819232
min            0.100000    0.000000
25%            0.300000    0.000000
50%            1.300000    1.000000
75%            1.800000    2.000000
max            2.500000    2.000000
```




```
Original Iris shape: (150, 5), After Outlier Removal: (142, 5)

Correlation Matrix:
                   sepal length (cm)  sepal width (cm)  petal length (cm)  \
sepal length (cm)           1.000000         -0.226237           0.870430
sepal width (cm)           -0.226237          1.000000          -0.508601
petal length (cm)           0.870430         -0.508601           1.000000
petal width (cm)            0.815823         -0.430027           0.961136
target                      0.779696         -0.491746           0.946469

                   petal width (cm)    target
sepal length (cm)          0.815823  0.779696
sepal width (cm)          -0.430027 -0.491746
petal length (cm)          0.961136  0.946469
petal width (cm)           1.000000  0.953940
target                     0.953940  1.000000
```



Question 2: WINE Dataset

For dataset 'Wines' use correlation matrix for feature selection and compare the various models. draw a bar chart of this summary. (use gini index as the criteria for classification)

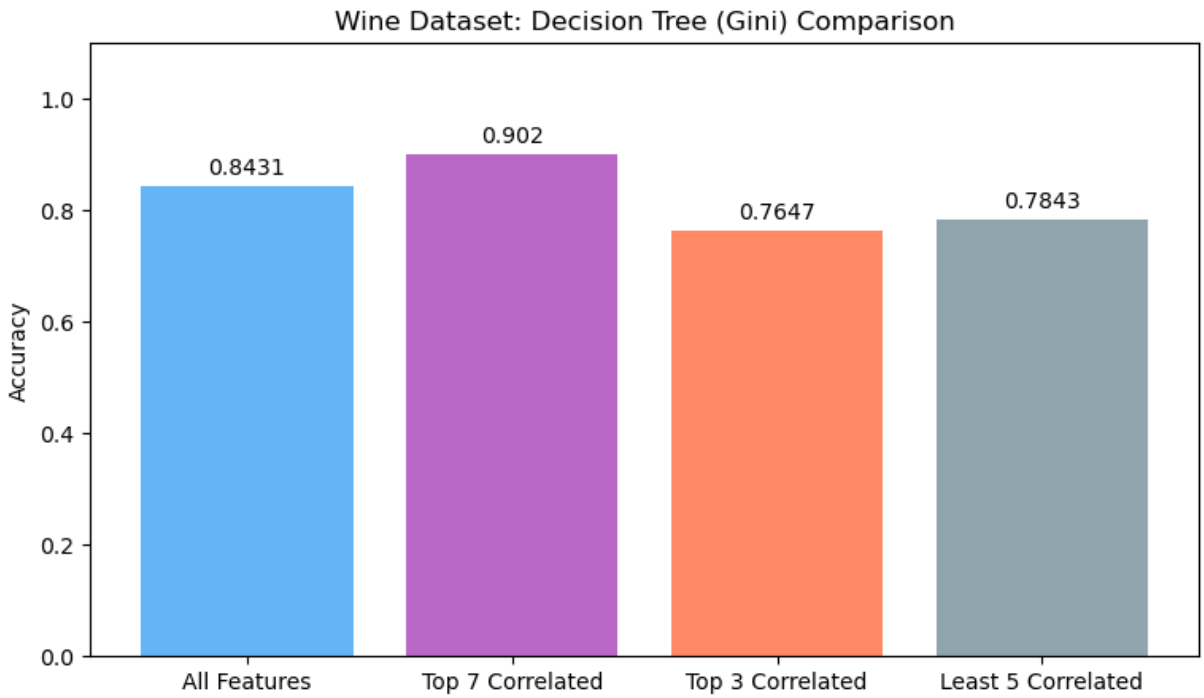
General Requirements: a) perform all the initial EDA analysis c) Remove Outlier in all the cases

In [2]:
```
print("=== WINE DATASET ===")
wine = load_wine()
df_wine = pd.DataFrame(wine.data, columns=wine.feature_names)
df_wine['target'] = wine.target

# a) Initial EDA analysis
print("\n--- Initial EDA: Info ---")
print(df_wine.info())
print("\n--- Initial EDA: Describe ---")
print(df_wine.describe())

# c) Remove Outliers
df_wine_clean = remove_outliers(df_wine, 'target')
print(f"\nOriginal Wine shape: {df_wine.shape}, After Outlier Removal: {df_wine_cle

# Feature Selection using Correlation Matrix
corr_matrix_wine = df_wine_clean.corr()
target_corr_wine = corr_matrix_wine['target'].drop('target').abs().sort_values(asce

X_wine = df_wine_clean.drop(columns=['target'])
y_wine = df_wine_clean['target']
```

```
# Comparing various models based on feature selection
wine_feature_sets = {
    'All Features': X_wine.columns.tolist(),
    'Top 7 Correlated': target_corr_wine.head(7).index.tolist(),
    'Top 3 Correlated': target_corr_wine.head(3).index.tolist(),
    'Least 5 Correlated': target_corr_wine.tail(5).index.tolist()
}

accuracies_wine = {}
for name, features in wine_feature_sets.items():
    X_subset = X_wine[features]
    X_train, X_test, y_train, y_test = train_test_split(X_subset, y_wine, test_size
    # Gini index as criteria
    clf = DecisionTreeClassifier(criterion='gini', random_state=42)
    clf.fit(X_train, y_train)
    accuracies_wine[name] = accuracy_score(y_test, clf.predict(X_test))

# Draw a bar chart of this summary
plt.figure(figsize=(9, 5))
bars = plt.bar(accuracies_wine.keys(), accuracies_wine.values(), color=['#64B5F6',
plt.ylabel('Accuracy')
plt.title('Wine Dataset: Decision Tree (Gini) Comparison')
plt.ylim(0, 1.1)
for bar in bars:
    yval = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2, yval + 0.02, round(yval, 4), ha='cent
plt.show()
```

```
=== WINE DATASET ===

--- Initial EDA: Info ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 178 entries, 0 to 177
Data columns (total 14 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   alcohol                       178 non-null    float64
 1   malic_acid                    178 non-null    float64
 2   ash                           178 non-null    float64
 3   alcalinity_of_ash             178 non-null    float64
 4   magnesium                     178 non-null    float64
 5   total_phenols                 178 non-null    float64
 6   flavanoids                    178 non-null    float64
 7   nonflavanoid_phenols          178 non-null    float64
 8   proanthocyanins               178 non-null    float64
 9   color_intensity               178 non-null    float64
 10  hue                           178 non-null    float64
 11  od280/od315_of_diluted_wines  178 non-null    float64
 12  proline                       178 non-null    float64
 13  target                        178 non-null    int64
dtypes: float64(13), int64(1)
memory usage: 19.6 KB
None

--- Initial EDA: Describe ---
          alcohol  malic_acid         ash  alcalinity_of_ash   magnesium  \
count  178.000000  178.000000  178.000000         178.000000  178.000000
mean    13.000618    2.336348    2.366517          19.494944   99.741573
std      0.811827    1.117146    0.274344           3.339564   14.282484
min     11.030000    0.740000    1.360000          10.600000   70.000000
25%     12.362500    1.602500    2.210000          17.200000   88.000000
50%     13.050000    1.865000    2.360000          19.500000   98.000000
75%     13.677500    3.082500    2.557500          21.500000  107.000000
max     14.830000    5.800000    3.230000          30.000000  162.000000

       total_phenols  flavanoids  nonflavanoid_phenols  proanthocyanins  \
count     178.000000  178.000000            178.000000       178.000000
mean        2.295112    2.029270              0.361854         1.590899
std         0.625851    0.998859              0.124453         0.572359
min         0.980000    0.340000              0.130000         0.410000
25%         1.742500    1.205000              0.270000         1.250000
50%         2.355000    2.135000              0.340000         1.555000
75%         2.800000    2.875000              0.437500         1.950000
max         3.880000    5.080000              0.660000         3.580000

       color_intensity         hue  od280/od315_of_diluted_wines      proline  \
count       178.000000  178.000000                    178.000000   178.000000
mean          5.058090    0.957449                      2.611685   746.893258
std           2.318286    0.228572                      0.709990   314.907474
min           1.280000    0.480000                      1.270000   278.000000
25%           3.220000    0.782500                      1.937500   500.500000
50%           4.690000    0.965000                      2.780000   673.500000
75%           6.200000    1.120000                      3.170000   985.000000
max          13.000000    1.710000                      4.000000  1680.000000
```

```
           target
count  178.000000
mean     0.938202
std      0.775035
min      0.000000
25%      0.000000
50%      1.000000
75%      2.000000
max      2.000000

Original Wine shape: (178, 14), After Outlier Removal: (169, 14)
```



Question 3: DIGIT Dataset

For dataset 'DIGIT' use PCA or SVD as dimensionality reduction techniques and compare 7 models (all feature, 30 components of PCA, 20 components of PCA, 15 components of PCA, 10 components of PCA, 5 components of PCA, 2 components of PCA). Summarise the results as a plot. (Use entropy as the criteria for classification).

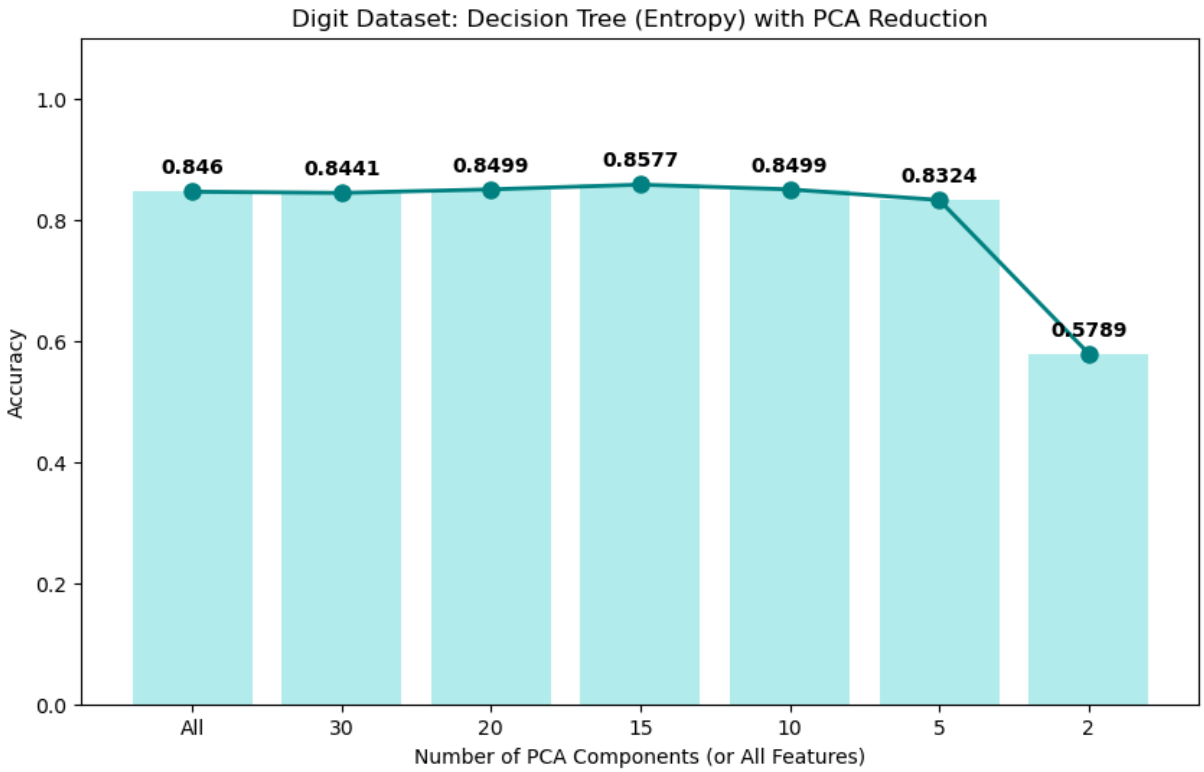
General Requirements: a) perform all the initial EDA analysis c) Remove Outlier in all the cases

In [3]:
```
print("=== DIGIT DATASET ===")
digits = load_digits()
df_digits = pd.DataFrame(digits.data)
df_digits['target'] = digits.target

# a) Initial EDA analysis
print("\n--- Initial EDA: Info ---")
print(df_digits.info())
print("\n--- Initial EDA: Data snippet ---")
```

```
print(df_digits.head())

# c) Remove Outliers
df_digits_clean = remove_outliers(df_digits, 'target')
print(f"\nOriginal Digits shape: {df_digits.shape}, After Outlier Removal: {df_digi

X_dig = df_digits_clean.drop(columns=['target'])
y_dig = df_digits_clean['target']

X_train_dig, X_test_dig, y_train_dig, y_test_dig = train_test_split(X_dig, y_dig, t

# Compare 7 models based on PCA components
pca_components = ['All', 30, 20, 15, 10, 5, 2]
accuracies_dig = {}

for comp in pca_components:
    if comp == 'All':
        X_tr, X_te = X_train_dig, X_test_dig
    else:
        pca = PCA(n_components=comp, random_state=42)
        X_tr = pca.fit_transform(X_train_dig)
        X_te = pca.transform(X_test_dig)

    # Entropy as criteria
    clf = DecisionTreeClassifier(criterion='entropy', random_state=42)
    clf.fit(X_tr, y_train_dig)
    y_pred = clf.predict(X_te)
    accuracies_dig[str(comp)] = accuracy_score(y_test_dig, y_pred)

# Summarise the results as a plot
plt.figure(figsize=(10, 6))
keys = list(accuracies_dig.keys())
values = list(accuracies_dig.values())

# Combining Line plot and Bar plot for better visualization of the trend
plt.plot(keys, values, marker='o', linestyle='-', color='teal', linewidth=2, marker
plt.bar(keys, values, alpha=0.3, color='c')
plt.ylabel('Accuracy')
plt.xlabel('Number of PCA Components (or All Features)')
plt.title('Digit Dataset: Decision Tree (Entropy) with PCA Reduction')
plt.ylim(0, 1.1)

for i, v in enumerate(values):
    plt.text(i, v + 0.03, str(round(v, 4)), ha='center', fontweight='bold')
plt.show()
```

```
=== DIGIT DATASET ===

--- Initial EDA: Info ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1797 entries, 0 to 1796
Data columns (total 65 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   0       1797 non-null   float64
 1   1       1797 non-null   float64
 2   2       1797 non-null   float64
 3   3       1797 non-null   float64
 4   4       1797 non-null   float64
 5   5       1797 non-null   float64
 6   6       1797 non-null   float64
 7   7       1797 non-null   float64
 8   8       1797 non-null   float64
 9   9       1797 non-null   float64
 10  10      1797 non-null   float64
 11  11      1797 non-null   float64
 12  12      1797 non-null   float64
 13  13      1797 non-null   float64
 14  14      1797 non-null   float64
 15  15      1797 non-null   float64
 16  16      1797 non-null   float64
 17  17      1797 non-null   float64
 18  18      1797 non-null   float64
 19  19      1797 non-null   float64
 20  20      1797 non-null   float64
 21  21      1797 non-null   float64
 22  22      1797 non-null   float64
 23  23      1797 non-null   float64
 24  24      1797 non-null   float64
 25  25      1797 non-null   float64
 26  26      1797 non-null   float64
 27  27      1797 non-null   float64
 28  28      1797 non-null   float64
 29  29      1797 non-null   float64
 30  30      1797 non-null   float64
 31  31      1797 non-null   float64
 32  32      1797 non-null   float64
 33  33      1797 non-null   float64
 34  34      1797 non-null   float64
 35  35      1797 non-null   float64
 36  36      1797 non-null   float64
 37  37      1797 non-null   float64
 38  38      1797 non-null   float64
 39  39      1797 non-null   float64
 40  40      1797 non-null   float64
 41  41      1797 non-null   float64
 42  42      1797 non-null   float64
 43  43      1797 non-null   float64
 44  44      1797 non-null   float64
 45  45      1797 non-null   float64
 46  46      1797 non-null   float64
 47  47      1797 non-null   float64
```

```
 48  48      1797 non-null   float64
 49  49      1797 non-null   float64
 50  50      1797 non-null   float64
 51  51      1797 non-null   float64
 52  52      1797 non-null   float64
 53  53      1797 non-null   float64
 54  54      1797 non-null   float64
 55  55      1797 non-null   float64
 56  56      1797 non-null   float64
 57  57      1797 non-null   float64
 58  58      1797 non-null   float64
 59  59      1797 non-null   float64
 60  60      1797 non-null   float64
 61  61      1797 non-null   float64
 62  62      1797 non-null   float64
 63  63      1797 non-null   float64
 64  target  1797 non-null   int64
dtypes: float64(64), int64(1)
memory usage: 912.7 KB
None

--- Initial EDA: Data snippet ---
     0    1    2     3     4     5    6    7    8    9  ...   55   56   57  \
0  0.0  0.0  5.0  13.0   9.0   1.0  0.0  0.0  0.0  0.0  ...  0.0  0.0  0.0
1  0.0  0.0  0.0  12.0  13.0   5.0  0.0  0.0  0.0  0.0  ...  0.0  0.0  0.0
2  0.0  0.0  0.0   4.0  15.0  12.0  0.0  0.0  0.0  0.0  ...  0.0  0.0  0.0
3  0.0  0.0  7.0  15.0  13.0   1.0  0.0  0.0  0.0  8.0  ...  0.0  0.0  0.0
4  0.0  0.0  0.0   1.0  11.0   0.0  0.0  0.0  0.0  0.0  ...  0.0  0.0  0.0

    58    59    60    61   62   63  target
0  6.0  13.0  10.0   0.0  0.0  0.0       0
1  0.0  11.0  16.0  10.0  0.0  0.0       1
2  0.0   3.0  11.0  16.0  9.0  0.0       2
3  7.0  13.0  13.0   9.0  0.0  0.0       3
4  0.0   2.0  16.0   4.0  0.0  0.0       4

[5 rows x 65 columns]

Original Digits shape: (1797, 65), After Outlier Removal: (1707, 65)
```


Digit Dataset: Decision Tree (Entropy) with PCA Reduction
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
0.846
0.8441
0.8499
0.8577
0.8499
0.8324
0.5789
All
30
20
15
10
5
2
Number of PCA Components (or All Features)